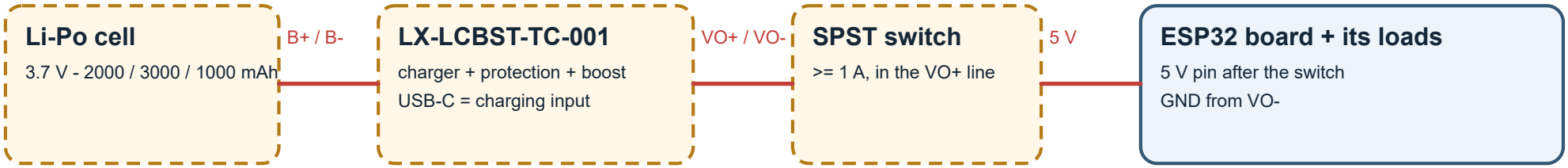
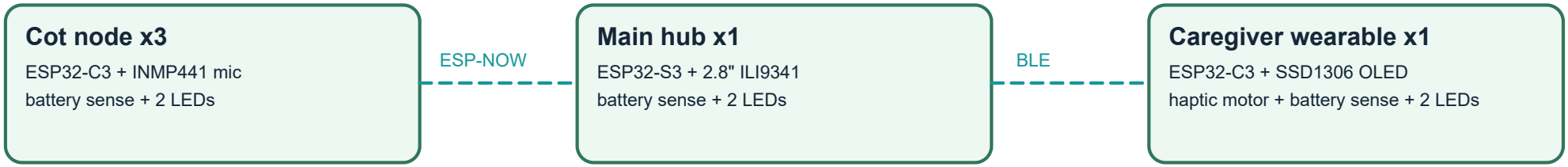
1 / CryNet system - power chain and links

Every unit has its own battery and its own ground. No wire ever joins two units.



One chain in all four units - only the loads differ:



Rules every unit follows:

1 - Switch OFF before charging. The cell still charges on B+/B-.

2 - Set the boost trimpot to 5.1 V with a meter before any board.

3 - Never power a board from USB and the boost at the same time.

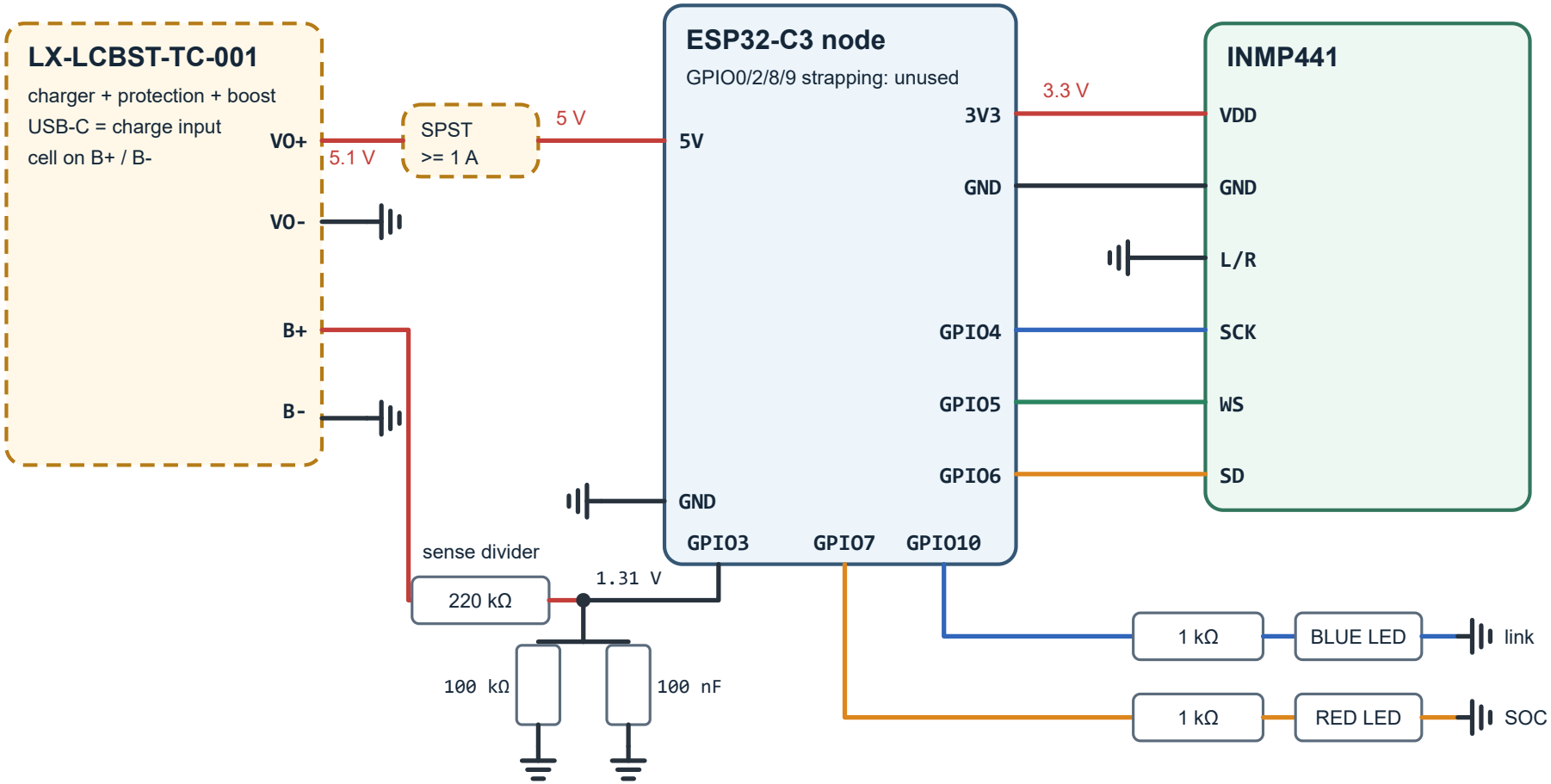
Dashed teal = wireless link, no wire. Solid = a real conductor.

> Proposed, bench-validation pending: the whole battery chain, the wearables and the LEDs. The node and hub signal wiring is the verified part.

No signal wire exists between the nodes and the hub - the link is wireless. Each unit is electrically independent; never join their batteries or grounds. Read with SYSTEM_WIRING.md sections 5 and 6.

2 / Cot node - complete wiring (one per node, x3)

ESP32-C3 SuperMini + INMP441 + LX-LCBST + switch + battery-sense divider + 2 LEDs



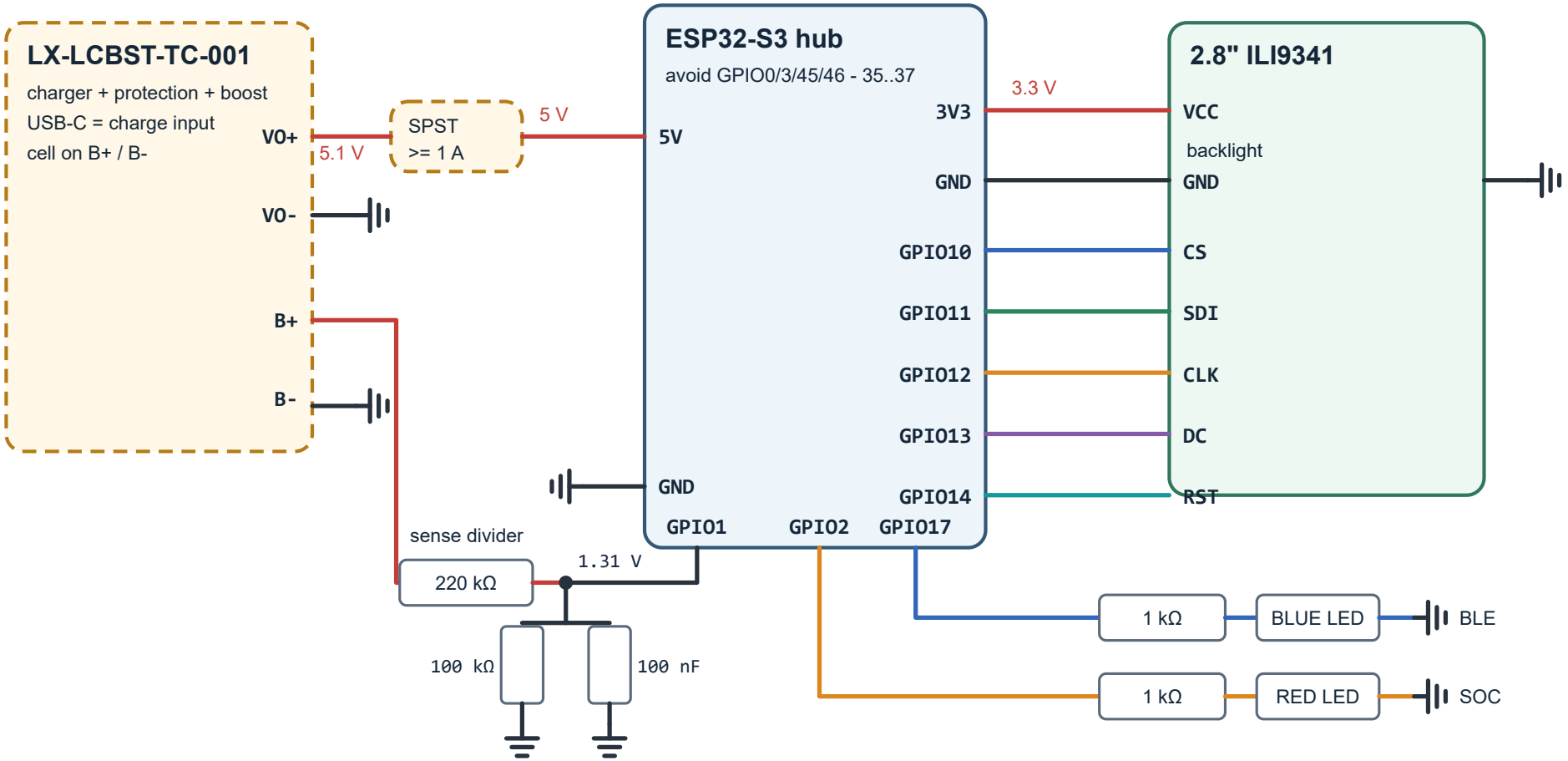
Verified on the live system: mic wiring (3V3 / GND / L/R→GND, GPIO4 SCK, GPIO5 WS, GPIO6 SD) and the node firmware. GPIO1 stays spare.

Proposed, needs bench validation: the whole battery chain, the divider and the LEDs. The divider reads about 13 uA. RED LED = SOC (off / slow / fast), BLUE LED = link (ACK-driven). Firmware for the divider and the LEDs does not exist yet - branch work.

Every ⏚ symbol is the same net: the module's VO- (battery negative through the module's protection FET). Wires join only where a dot is drawn. Pin names, not header positions. Repeat identically for nodes 2 and 3.

3 / Main hub - complete wiring (x1)

ESP32-S3-DevKitC-1 N16R8 + 2.8" ILI9341 + LX-LCBST + switch + battery-sense divider + 2 LEDs



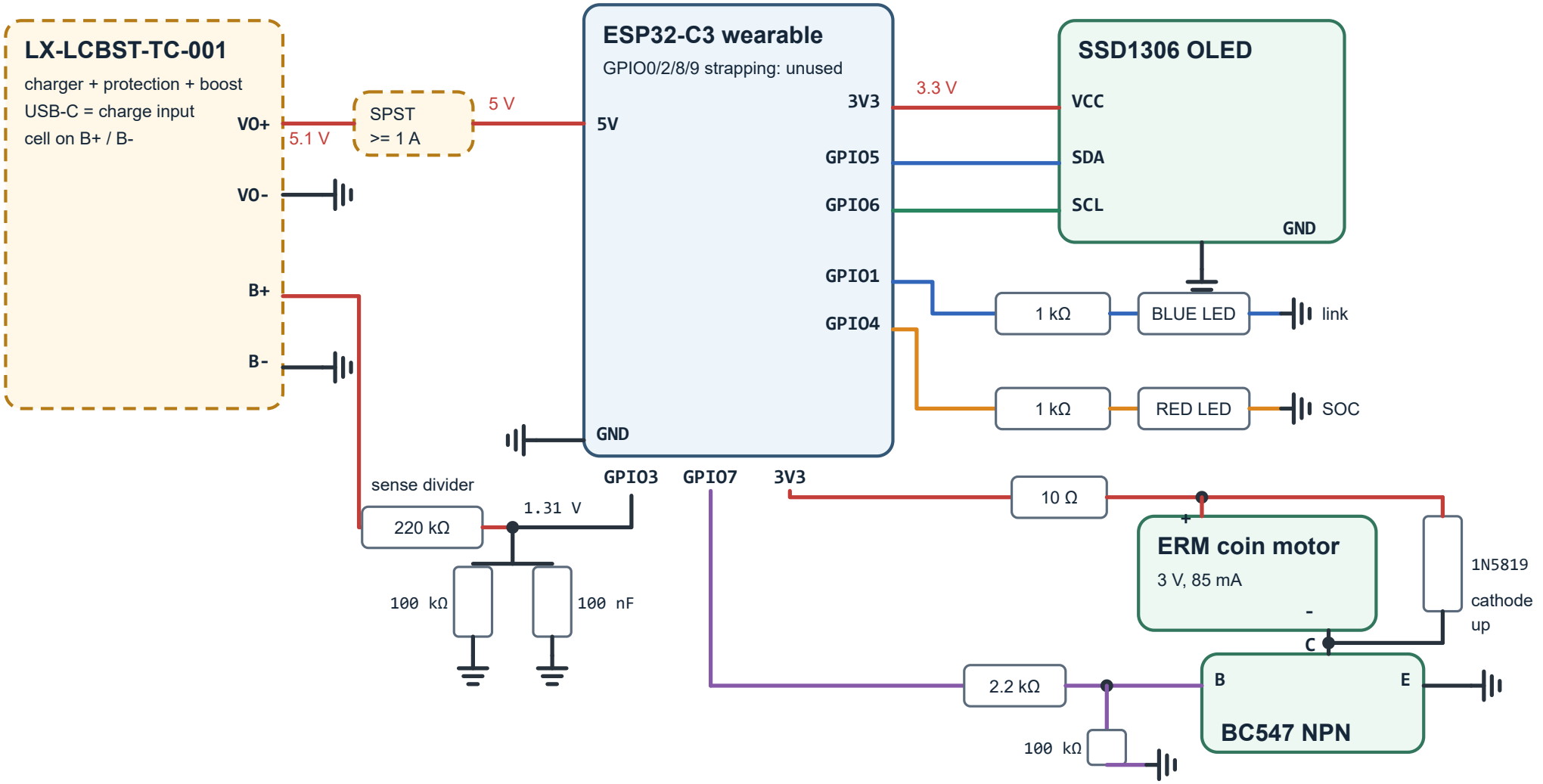
Verified on the live system: TFT signals (CS 10, MOSI 11, SCK 12, DC 13, RST 14, BL 4, VCC 3V3, GND) and the hub firmware.

Proposed, needs bench validation: the battery chain, the divider on GPIO1 (ADC1_CH0), BLUE LED on GPIO2 (BLE client connected) and RED LED on GPIO17 (SOC). The hub's peak current with TFT + radios is unmeasured - confirm the LX-LCBST holds 5.1 V during bursts before trusting it here.

GPIO35-37 are the Octal PSRAM bus - never wire them. The TFT wiring is the recorded working one for the existing module. Every ⏚ symbol is the same net: the module's VO-.

4 / Caregiver wearable - complete wiring (x1)

ESP32-C3 SuperMini + SSD1306 OLED + LX-LCBST + switch + battery-sense divider + 2 LEDs + BC547 haptic drive

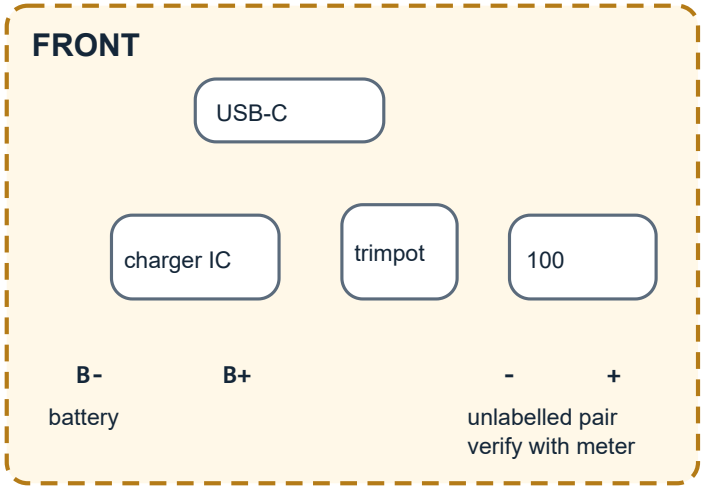


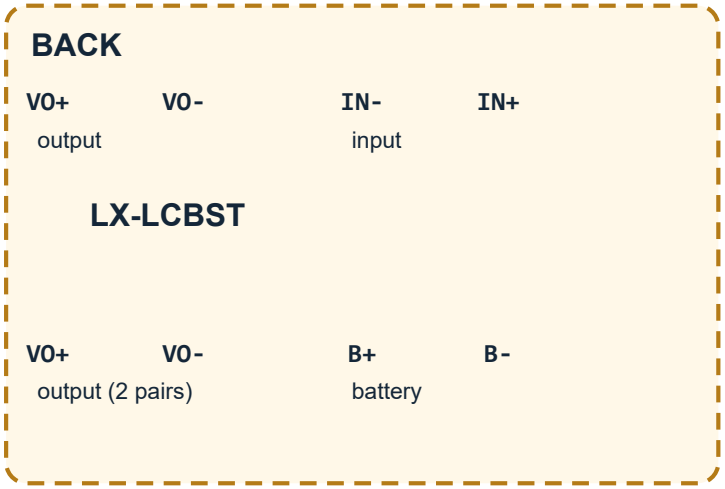
Divider confirmed: 220 kΩ from B+ and 100 kΩ to ground give 4.2 V × 100/320 = 1.31 V at the ADC (the 100 nF filters it), 13 uA of divider current; the node firmware already uses this ratio (reading × 3.2). Motor: the 10 Ω caps the stall current at 3.3 V ÷ (R + 10 Ω) - about 69 mA for a 37 Ω coin ERM, inside the BC547's 100 mA; a lower-impedance motor needs 22 Ω or a 2N2222A. The firmware matches this page: GPIO7 drives the switch in 250 ms pulses, one buzz per cry. The two LEDs and the divider are on the board, but the wearable firmware does not read the ADC or light the LEDs yet.

Every ⏚ symbol is the same net; wires join only where a dot is drawn. The BC547 emitter is the motor's ground, and the flyback sits across the motor with the cathode up.

5 / LX-LCBST-TC-001 - pad map and bench procedure

From the user's front-and-back photograph - Chinese silk text translated





Bench procedure, before any ESP32 is connected:

1 - Cell on B+/B-: turn the trimpot until VO+/VO- reads 5.1 V on a meter, no load.

2 - USB-C in: measure the charge current with a series meter - the cell must allow it.

3 - Confirm the charge ends at 4.2 V.

4 - Switch off: confirm the load is dead and measure the battery drain (boost quiescent unknown).

5 - Run one node and compare it with the USB baseline.

No T pad on this board - the listing's T-shorting notes belong to other variants. Never short anything.

Unknown until measured: the actual charge current (the program resistor is unreadable in the photo), the boost quiescent drain when off, the protection thresholds, and what the front - / + pair connects to. Buy ONE module for validation; only then buy the remaining three (plus the wearable's, pending the same check).